Multimodal Attendance & Engagement Agent

Pre-requisites

Epic | Complexity: Medium | Duration: 1h 0m

Description

Pre-requisites are the essential tools, software, or knowledge required before starting a project.
They ensure the project can be set up, run, and developed smoothly without issues.

Resources

No resources available

Project Progress

50%

Add Demo Link | Add Github Link

- Pre-requisites 50%
  - Pre-requisites
- Project Workflow 50%
- EPIC 1: Requirements Specification and Environment Setup 50%
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- EPIC 3: Audio Module Development 50%
- EPIC 4: AI IMPLEMENTATION 50%
- EPIC 5 : Analytics and Recommendation Engine 50%
- EPIC 6 : Interface, Testing, and Deployment 50%
- Conclusion: 50%

Multimodal Attendance & Engagement Agent

Pre-requisites

Story | Duration: 1h 0m

In Progress

AR Assigned to **Arpan Mishra**

Description

Development Knowledge

- Python programming fundamentals
- Computer Vision concepts (OpenCV, MediaPipe)
- Audio signal processing basics
- Understanding of agent-based AI systems
- Familiarity with Streamlit dashboards

Tools & Frameworks

- **Streamlit** → https://streamlit.io
- **OpenCV** → https://opencv.org
- **MediaPipe** → https://developers.google.com/mediapipe
- **Librosa** → https://librosa.org
- **Python 3.9+** → https://python.org
- **Visual Studio Code** → https://code.visualstudio.com

Execution Environment

- Offline / Local execution
- Webcam or video file input
- Microphone or audio file input (optional)

Resources

No resources available

Project Details

Project Progress 50%

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  - Pre-requisites
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- EPIC 5 : Analytics and Recommendation Engine 50%
- EPIC 6 : Interface, Testing, and Deployment 50%
- Conclusion: 50%

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Multimodal Attendance & Engagement Agent

workflow

Story | Duration: 1h 0m

In Progress

AR Assigned to **Arpan Mishra**

Description

Milestone 1: Requirements Specification and Environment Setup

This milestone focuses on defining functional requirements, selecting libraries, and preparing the development environment.

Activity 1.1: Create requirements.txt
Activity 1.2: Install required libraries and set up virtual environment

Milestone 2: Vision Module Development

This milestone implements face detection, tracking, and presence duration logic to enable reliable attendance marking.

Activity 2.1: Face detection and tracking
Activity 2.2: Presence-based attendance logic

Milestone 3: Audio Module Development

This milestone introduces audio processing for noise and speech activity detection.

Activity 3.1: Audio signal extraction
Activity 3.2: Noise and speech activity analysis

Milestone 4: Multimodal Agent Implementation

This milestone focuses on aggregating signals from vision and audio modules and applying rule-based reasoning.

Activity 4.1: Multimodal signal fusion
Activity 4.2: Engagement scoring logic

Milestone 5: Analytics and Recommendation Engine

This milestone generates instructor suggestions and student session analytics based on engagement trends.

Activity 5.1: Teaching suggestion logic
Activity 5.2: Student analytics generation

Milestone 6: Interface, Testing, and Deployment

This milestone validates system performance and provides optional visualization via a dashboard.

Activity 6.1: Run core application
Activity 6.2: Launch optional Streamlit dashboard
Activity 6.3: Validate outputs and edge cases

Project Details

Project Progress 50%

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- Pre-requisites 50%
- Project Workflow 50%
  - workflow
- EPIC 1: Requirements Specification and Environment Setup 50%
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- EPIC 6 : Interface, Testing, and Deployment 50%
- Conclusion: 50%

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Activity 1.1: Create requirements.txt

Story
Duration: 1h 0m

In Progress

AR Assigned to **Arpan Mishra**

Description

Approach

The first step involved identifying all the software dependencies required for implementing the core functionalities of the system. Since the project involves computer vision, audio signal processing, backend logic, and full-stack integration, libraries were selected to support each of these components. Priority was given to open-source, well-documented, and widely adopted libraries to ensure maintainability and ease of installation.

Process

1. Analyzed the project features such as face detection, engagement analysis, audio processing, backend API handling, and report generation.
2. Identified libraries required for each functional module:
   - Vision processing
   - Audio analysis
   - Backend and API handling
   - Data storage and analytics
3. Verified compatibility of selected libraries with the targeted Python version.
4. Documented all required dependencies and their versions in a requirements.txt file.
5. Reviewed the dependency list to eliminate redundant or conflicting libraries.

Technical Details

The requirements.txt file contains dependencies grouped into the following categories:

- **Computer Vision Libraries**
  Used for face detection, tracking, and engagement analysis.
- **Audio Processing Libraries**
  Used for extracting audio features, noise levels, and speech activity cues.
- **Backend and API Libraries**
  Used for handling requests, managing sessions, and serving data to the frontend.
- **Data Processing Libraries**
  r analytics, report generation, and structured data handling.

uration enables automated installation of all required packages, ensuring environment reproducibility.

backend > requirements.txt

```
fastapi
uvicorn
opencv-python
mediapipe
numpy
pandas
python-multipart
websockets
sqlalchemy
librosa
soundfile
groq
deep-sort-realtime
python-dotenv
```

Project Details

Project Progress 50%

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Pre-requisites 50%
Project Workflow 50%
EPIC 1: Requirements Specification and Environment Setup 50%
Activity 1.1: Create requirements.txt
Activity 1.2: Install Required Libraries and Set Up Virtual Environment
EPIC 2: Vision Module Development 50%
EPIC 3: Audio Module Development 50%
EPIC 4: AI IMPLEMENTATION 50%
EPIC 5 : Analytics and Recommendation Engine 50%
EPIC 6 : Interface, Testing, and Deployment 50%
Conclusion: 50%

Multimodal Attendance & Engagement Agent

Activity 1.2: Install Required Libraries and Set Up Virtual Environment

Story | Duration: 1h 0m

In Progress

AR Assigned to **Arpan Mishra**

Description

Approach

To ensure isolation from system-level Python packages and prevent version conflicts, a dedicated virtual environment was created for the project. This approach improves reproducibility and aligns with best practices for professional software development.

Process

1. Verified Python installation and version compatibility.
2. Created a virtual environment specific to the project.
3. Activated the virtual environment before installing dependencies.
4. Installed all required libraries using the requirements.txt file.
5. Validated successful installation by importing key libraries and running a basic test script.

Technical Details

- Python virtual environment was used to manage dependencies.
- All libraries were installed using the package manager through a single command referencing requirements.txt.
- Successful setup was confirmed by launching the backend server and verifying module imports.

This setup ensures that the development environment is portable, stable, and aligned with industry standards.

Resources

No resources available

Project Details

Project Progress 50%

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Pre-requisites 50%
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EPIC 1: Requirements Specification and Environment Setup 50%
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EPIC 5 : Analytics and Recommendation Engine 50%
EPIC 6 : Interface, Testing, and Deployment 50%
Conclusion: 50%

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Multimodal Attendance & Engagement Agent

Description

In this activity, face detection and tracking mechanisms were implemented to identify and monitor participants throughout the session. Video input was captured either from a live webcam feed or a pre-recorded classroom video. Each frame was processed sequentially to detect human faces using computer vision techniques.

Once detected, each face was assigned a unique identifier to maintain identity consistency across frames. Tracking logic was applied to associate detected faces across consecutive frames based on spatial location and movement patterns. This approach ensures that participants are continuously tracked even when slight head movements, orientation changes, or brief occlusions occur.

The face detection and tracking process generates structured data containing participant identifiers, timestamps, and positional information. This continuous tracking capability serves as the foundation for calculating presence duration and further engagement analysis.

backend > core > face_detection.py

```python
import cv2
import os

class FaceDetector:
    def __init__(self):
        # Use OpenCV's built-in Haarcascade for face detection
        # This is more robust across Python versions than MediaPipe
        haarcascade_path = cv2.data.haarcascades + 'haarcascade_frontalface_default.xml'
        self.face_cascade = cv2.CascadeClassifier(haarcascade_path)
        if self.face_cascade.empty():
            print("Error: Could not load face cascade xml")

    def process_frame(self, frame):
        # Convert to grayscale for detection
        gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

        # Detect faces
        # scaleFactor=1.1, minNeighbors=5, minSize=(30, 30)
        faces = self.face_cascade.detectMultiScale(gray, 1.1, 5, minSize=(30, 30))

        detections = []
        for (x, y, w, h) in faces:
            # Format to match previous output: bbox (x, y, w, h), score (1.0 placeholder)
            detections.append({"bbox": (int(x), int(y), int(w), int(h)), "score": 1.0})
```

backend > core > camera_service.py

```python
import cv2
import time
import asyncio
from fastapi import WebSocket

class CameraService:
    def __init__(self, camera_id=0):
        self.camera_id = camera_id
        self.cap = None
        self.is_running = False

    def start(self):
        if self.cap is None:
            self.cap = cv2.VideoCapture(self.camera_id)
            self.is_running = True

    def stop(self):
        if self.cap:
            self.cap.release()
            self.cap = None
            self.is_running = False

    def get_frame(self):
        if self.is_running and self.cap:
            ret, frame = self.cap.read()
            if ret:
                return frame
```

Add Demo Link | Add Github Link

- Pre-requisites 50%
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  - Activity 2.2: Presence-Based Attendance Logic
- EPIC 3: Audio Module Development 50%
- EPIC 4: AI IMPLEMENTATION 50%
- EPIC 5 : Analytics and Recommendation Engine 50%
- EPIC 6 : Interface, Testing, and Deployment 50%
- Conclusion: 50%

Multimodal Attendance & Engagement Agent

AR Assigned to **Arpan Mishra**

Description

This activity focuses on translating face tracking data into meaningful attendance records. Instead of marking attendance based on a single frame detection, the system uses **presence-based logic**, where attendance is confirmed only when a participant remains visible for a minimum duration during the session.

Presence duration was calculated using timestamps collected during the tracking phase. Intermittent appearances were accumulated to ensure fairness and to prevent false attendance marking caused by brief or accidental detections. Once the cumulative presence exceeded a predefined threshold, the participant was marked as present.

dance results were stored in structured logs, including participant identifiers and presence duration. This approach curacy, reduces false positives, and provides reliable attendance records suitable for classroom analytics and

backend > core > face_recognition.py

```python
from deepface import DeepFace
import cv2
import numpy as np
import os

class FaceRecognizer:
    def __init__(self, db_path="student_db"):
        self.db_path = db_path
        if not os.path.exists(db_path):
            os.makedirs(db_path)

    def register_student(self, img_path, student_id):
        # In a real app, we'd store embeddings.
        # Deepface.find uses a folder of images as DB.
        pass

    def recognize(self, frame):
        try:
            # Save frame temporarily or pass numpy array directly if supported (Deepface supports numpy)
            # Deepface.find requires a db_path with images.
            # If no DB yet, we can't recognize.
            if not os.listdir(self.db_path):
                return []

            # This is computing intensive, so we should run it sparingly
            dfs = DeepFace.find(img_path=frame, db_path=self.db_path, silent=True, enforce_detection=False)
```

Resources

No resources available

Project Progress

50%

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- Conclusion: 50%

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Multimodal Attendance & Engagement Agent

AR Assigned to **Arpan Mishra**

Description

In this activity, audio input was captured from a microphone or extracted from recorded video files associated with classroom sessions. The raw audio signal was processed to convert it into a digital waveform suitable for analysis. Sampling rate normalization and channel standardization were applied to ensure consistent processing across different audio sources.

The extracted audio signal was segmented into fixed-duration windows to enable time-based analysis aligned with video ▲ iese audio segments form the base input for noise and speech activity detection in subsequent stages.

▼ ssed audio data was stored in structured formats, including timestamps and amplitude information, to facilitate ed multimodal analysis.

backend > core > audio_analysis.py

```python
import librosa
import numpy as np
import os

class AudioAnalyzer:
    def __init__(self, sample_rate=16000):
        self.sample_rate = sample_rate

    def analyze_audio_chunk(self, audio_data):
        """
        Analyzes raw audio bytes/array:
        1. Noise level (decibels)
        2. Speech vs silence ratio
        3. Sound activity patterns

        Returns engagement metrics
        """
        try:
            # Convert bytes to float32 numpy array if needed
            if isinstance(audio_data, bytes):
                # Assuming 16-bit PCM
                y = np.frombuffer(audio_data, dtype=np.int16).astype(np.float32) / 32768.0
            else:
                y = audio_data

            if len(y) == 0:
                return None
```

Resources

No resources available

Project Progress

50%

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  - Activity 3.2:Noise and Speech Activity Analysis
- EPIC 4: AI IMPLEMENTATION 50%
- EPIC 5 : Analytics and Recommendation Engine 50%
- EPIC 6 : Interface, Testing, and Deployment 50%
- Conclusion: 50%

Multimodal Attendance & Engagement Agent

AR Assigned to **Arpan Mishra**

Description

This activity focuses on analyzing the extracted audio signals to identify ambient noise levels and speech activity patterns. Audio features such as energy levels, amplitude variation, and short-time signal characteristics were computed for each segment. These features were used to differentiate between periods of silence, background noise, and active speech.

Noise level trends provide insights into classroom environment conditions, while speech activity detection indicates interaction intensity and participation frequency. The results of this analysis were aggregated over time to produce session-level audio metrics.

ated audio insights were structured in a format suitable for fusion with visual engagement data, enabling the l agent to reason about overall session dynamics.

frontend > src > components > AudioCapture.jsx > ...

```jsx
import React, { useEffect, useRef, useState } from 'react';

const AudioCapture = ({ active, onMetrics }) => {
    const [isRecording, setIsRecording] = useState(false);
    const wsRef = useRef(null);
    const audioContextRef = useRef(null);
    const processorRef = useRef(null);
    const inputRef = useRef(null);

    useEffect(() => {
        if (active) {
            startCapture();
        } else {
            stopCapture();
        }
        return () => stopCapture();
    }, [active]);

    const startCapture = async () => {
        try {
            const stream = await navigator.mediaDevices.getUserMedia({ audio: true });
            setIsRecording(true);

            // Connect WS
            const ws = new WebSocket('ws://localhost:8000/ws/audio');
            wsRef.current = ws;
```

Resources

No resources available

Project Progress

50%

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- EPIC 5 : Analytics and Recommendation Engine 50%
- EPIC 6 : Interface, Testing, and Deployment 50%
- Conclusion: 50%

Multimodal Attendance & Engagement Agent

Activity 4.1: Multimodal Agent Implementation

Story | Duration: 1h 0m

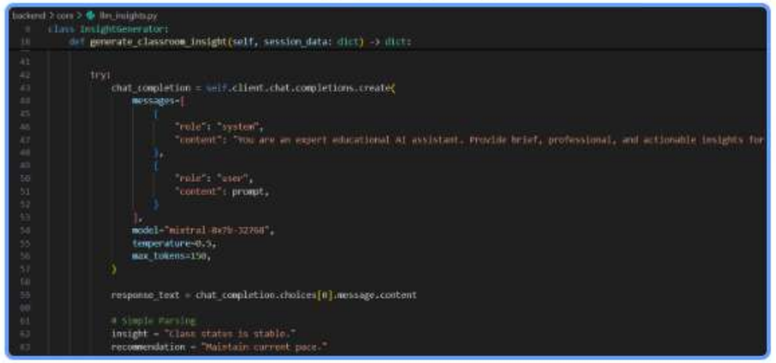
In Progress

AR Assigned to **Arpan Mishra**

Description

This milestone focuses on the implementation of the **Multimodal Agent**, which serves as the core intelligence layer of the system. The agent integrates outputs from both the Vision Module and the Audio Module to form a unified understanding of classroom behavior. Rather than relying on isolated signals, the agent applies **rule-based reasoning** to interpret combined multimodal data and derive meaningful engagement insights.

The primary goal of this milestone is to ensure that engagement estimation and session analysis are based on **contextual, interpretable, and explainable logic**, making the system suitable for real-world educational environments.



```
backend > core > llm_insights.py
class InsightGenerator:
    def generate_classroom_insight(self, session_data: dict) -> dict:

        try:
            chat_completion = self.client.chat.completions.create(
                messages=[
                    {
                        "role": "system",
                        "content": "You are an expert educational AI assistant. Provide brief, professional, and actionable insights for
                    },
                    {
                        "role": "user",
                        "content": prompt,
                    }
                ],
                model="mistral-8x7b-32768",
                temperature=0.5,
                max_tokens=150,
            )

            response_text = chat_completion.choices[0].message.content

            # Simple Parsing
            insight = "Class status is stable."
            recommendation = "Maintain current pace."
```

Resources

No resources available

Project Details

Project Progress 50%

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- Conclusion: 50%

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Multimodal Attendance & Engagement Agent

Activity 4.2: Multimodal Signal Fusion

Story | Duration: 1h 0m

In Progress

AR Assigned to **Arpan Mishra**

Description

In this activity, signals generated by the Vision Module and Audio Module were synchronized and aggregated into a unified data representation. Visual indicators such as presence duration, movement patterns, and inactivity periods were aligned with corresponding audio indicators including noise levels and speech activity using timestamp-based synchronization.

The fusion process ensures that both modalities contribute proportionally to engagement understanding. By correlating visual attention cues with audio interaction levels, the system captures a more accurate picture of classroom dynamics than single-nalysis. This combined representation forms the input for higher-level reasoning and scoring mechanisms.

data structure was designed to support extensibility, allowing additional modalities or features to be integrated in ancements.

backend > core > ai_suggestions.py

```python
class AISuggestionEngine:
    def generate_suggestions(self, session_data: dict):
        """

        try:
            completion = self.client.chat.completions.create(
                messages=[
                    {"role": "system", "content": "You are an expert AI teaching assistant."},
                    {"role": "user", "content": prompt}
                ],
                model="llama3-8b-8192",
                temperature=0.7,
                max_tokens=200
            )

            response = completion.choices[0].message.content
            return self._parse_response(response)

        except Exception as e:
            print(f"AI Suggestion Error: {e}")
            return self._fallback_suggestions(session_data)
```

Resources

No resources available

Project Details

Project Progress 50%

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- EPIC 6 : Interface, Testing, and Deployment 50%
- Conclusion: 50%

Multimodal Attendance & Engagement Agent

Activity 4.3: Engagement Scoring Logic

Story | Duration: 1h 0m

In Progress

AR Assigned to **Arpan Mishra**

Description

This activity focuses on transforming fused multimodal signals into interpretable **engagement scores**. Rule-based logic and predefined thresholds were applied to assess engagement levels across different time segments of the session. Factors such as sustained visual presence, movement frequency, inactivity duration, and speech activity intensity were collectively considered.

Engagement scores were computed at regular intervals to generate a temporal engagement profile rather than a single aggregated value. This approach allows identification of attention fluctuations, peak engagement periods, and extended disengagement phases. The scoring logic was designed to be transparent and adjustable, enabling fine-tuning based on classroom requirements.

The resulting engagement scores are used downstream to generate teaching suggestions, student analytics, and session summaries.

Resources

No resources available

Project Details

Project Progress

50%

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Pre-requisites 50%
Project Workflow 50%
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- Activity 4.1: Multimodal Agent Implementation
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- Activity 4.3: Engagement Scoring Logic

EPIC 5 : Analytics and Recommendation Engine 50%
EPIC 6 : Interface, Testing, and Deployment 50%
Conclusion: 50%

Multimodal Attendance & Engagement Agent

Description

In this activity, instructor-facing teaching suggestions were generated using engagement trends derived from the multimodal agent. Temporal engagement profiles were analyzed to detect patterns such as sustained disengagement, attention drops, high-interaction periods, and noisy classroom conditions.

Rule-based logic w7as applied to convert these patterns into **context-aware recommendations**. Examples include suggesting interactive activities during extended low-engagement periods, highlighting effective teaching segments associated with peak engagement, and flagging classroom management cues during high noise intervals.

The teaching suggestion mechanism is designed to support instructional decision-making rather than replace it. All suggestions are grounded in observable data trends and presented in a non-prescriptive manner, ensuring transparency and trust.

backend > core > teacher_profiles.py

```python
from sqlalchemy.orm import Session
from .database import SessionLocal, Session as SessionModel
import random

class TeacherProfileService:
    def get_profile(self, teacher_id: str):
        # Mock calculation - in production, aggregate real session data
        # Real logic: fetch average attention from all sessions by this teacher

        return {
            "teacher_id": teacher_id,
            "effectiveness_score": 87.5,
            "trend": 12.0, # +12%
            "class_history": [
                {"name": "CS101 (Today)", "engagement": 85},
                {"name": "CS101 (Yesterday)", "engagement": 78},
                {"name": "Math Workshop", "engagement": 92}
            ],
            "student_trends": [
                {"name": "Priya", "change": 15, "status": "improved"},
                {"name": "Kumar", "change": 8, "status": "improved"},
                {"name": "Asha", "change": -5, "status": "declining"}
            ],
            "strengths": [
                "High engagement sessions (85%+)",
                "Emotion diversity management"
```

backend > core > recommendations_engine.py

```python
class RecommendationsEngine:
    def generate_realtime_recommendations(self, session_status: dict):
        """
        Rule-based recommendations for immediate feedback.
        Input: Session status dict (people, emotions, attention)
        Output: List of { type: 'alert'|'warning'|'info', message: str }
        """
        recs = []

        # 1. Low Attention Alert
        avg_att = 0
        people = session_status.get('people', [])
        if people:
            avg_att = sum(p['attention'] for p in people) / len(people)

            # Individual check
            for p in people:
                if p['attention'] < 40:
                    recs.append({
                        'type': 'alert',
                        'message': f"Student {p['id']} is distracted ({p['attention']}%)."
                    })

        # 2. Global Attention Warning
        if len(people) > 2 and avg_att < 60:
            recs.append({
```

Resources

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  - Activity 5.1: Teaching Suggestion Logic
  - Activity 5.2 : Student Session Analytics Generation
- EPIC 6 : Interface, Testing, and Deployment 50%
- Conclusion: 50%

Multimodal Attendance & Engagement Agent

AR Assigned to **Arpan Mishra**

Description

This activity focuses on generating student-facing analytics that summarize individual participation and engagement patterns during a session. Using attendance records and engagement scores, the system produces structured session summaries that include attendance duration, engagement consistency, and notable activity trends.

Student analytics are presented in a reflective format, allowing learners to understand their participation without assigning grades or evaluations. Engagement timelines help students identify periods of high focus and distraction, encouraging self-ent and accountability.

ics were structured to ensure privacy, avoiding peer comparisons or ranking, and are suitable for dashboard on or report generation.

backend > core > analytics_service.py

```python
from sqlalchemy.orm import Session
from sqlalchemy import func
from .database import SessionLocal, Session as SessionModel, SessionPerson, PersonMetric

class AnalyticsService:
    def get_session_trends(self, session_id: int):
        db = SessionLocal()
        try:
            # Aggregate attention over time (minute buckets)
            # SQLite specific time grouping
            # For simplicity, we just return all metrics and let frontend downsample or bin

            # Fetch last 1000 points to avoid overload
            metrics = db.query(PersonMetric)\
                .filter(PersonMetric.session_id == session_id)\
                .order_by(PersonMetric.timestamp.asc())\
                .all()

            # Group by timestamp (seconds)
            timeline = {}
            for m in metrics:
                ts_str = m.timestamp.strftime("%H:%M:%S")
                if ts_str not in timeline:
                    timeline[ts_str] = {'att': [], 'emo': {}}
```

Resources

No resources available

Project Progress

50%

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- EPIC 5 : Analytics and Recommendation Engine 50%
  - Activity 5.1: Teaching Suggestion Logic
  - Activity 5.2 : Student Session Analytics Generation
- EPIC 6 : Interface, Testing, and Deployment 50%
- Conclusion: 50%

Multimodal Attendance & Engagement Agent

Activity 6.1: Run Core Application

Story | Duration: 1h 0m

In Progress

AR Assigned to **Arpan Mishra**

Description

In this activity, the complete application pipeline was executed in a local environment to verify end-to-end functionality. The system was tested using live webcam input and pre-recorded classroom videos to validate attendance detection, engagement analysis, and analytics generation.

The execution confirmed successful data flow across all modules, including vision processing, audio analysis, multimodal fusion, engagement scoring, and analytics generation. This step ensured that the core application performs reliably under normal operating conditions.

Resources

No resources available

Project Details

Project Progress 50%

Add Demo Link | Add Github Link

Pre-requisites 50%
Project Workflow 50%
EPIC 1: Requirements Specification and Environment Setup 50%
EPIC 2: Vision Module Development 50%
EPIC 3: Audio Module Development 50%
EPIC 4: AI IMPLEMENTATION 50%
EPIC 5 : Analytics and Recommendation Engine 50%
EPIC 6 : Interface, Testing, and Deployment 50%
Activity 6.1: Run Core Application
Activity 6.2 : Launch Optional Streamlit Dashboard
Activity 6.3 : Validate Outputs and Edge Cases
Conclusion: 50%

Multimodal Attendance & Engagement Agent

Activity 6.2 : Launch Optional Streamlit Dashboard

Story | Duration: 1h 0m

In Progress

AR Assigned to **Arpan Mishra**

Description

In this activity, an optional Streamlit-based dashboard was used to visualize system outputs. The dashboard displayed attendance records, engagement timelines, instructor suggestions, and student session analytics in an interactive format.

The visualization layer was designed to improve interpretability of system outputs rather than to serve as a mandatory interface. Charts, tables, and expandable sections allowed users to explore session data and insights effectively.

This activity demonstrated the system's capability to present complex multimodal analytics in a user-friendly manner.

Resources

No resources available

Project Details

Project Progress 50%

Add Demo Link | Add Github Link

Pre-requisites 50%
Project Workflow 50%
EPIC 1: Requirements Specification and Environment Setup 50%
EPIC 2: Vision Module Development 50%
EPIC 3: Audio Module Development 50%
EPIC 4: AI IMPLEMENTATION 50%
EPIC 5 : Analytics and Recommendation Engine 50%
EPIC 6 : Interface, Testing, and Deployment 50%
Activity 6.1: Run Core Application
Activity 6.2 : Launch Optional Streamlit Dashboard
Activity 6.3 : Validate Outputs and Edge Cases
Conclusion: 50%

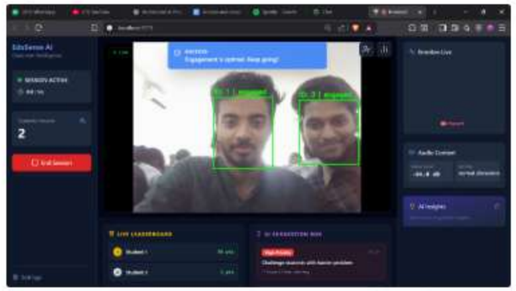
Description

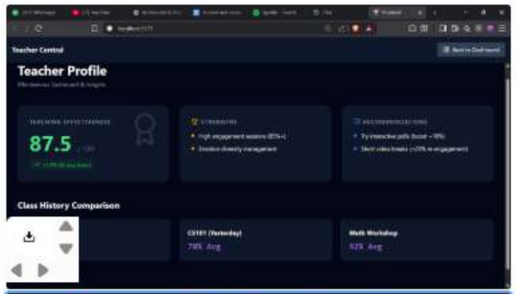
This activity focused on validating the accuracy and robustness of the system under various real-world conditions. Edge cases such as fluctuating lighting, background noise, partial occlusions, short participant appearances, and intermittent audio input were tested.

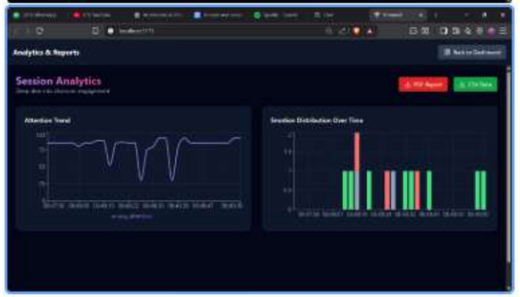
System outputs were reviewed to ensure that attendance marking remained accurate, engagement scores were stable, and analytics outputs were interpretable. Any inconsistencies were analyzed and mitigated through parameter adjustments and logic refinement.

This validation confirmed the system's readiness for real-world deployment in classroom and training environments.

Outputs:







Add Demo Link | Add Github Link

- Pre-requisites 50%
- Project Workflow 50%
- EPIC 1: Requirements Specification and Environment Setup 50%
- EPIC 2: Vision Module Development 50%
- EPIC 3: Audio Module Development 50%
- EPIC 4: AI IMPLEMENTATION 50%
- EPIC 5 : Analytics and Recommendation Engine 50%
- EPIC 6 : Interface, Testing, and Deployment 50%
  - Activity 6.1: Run Core Application
  - Activity 6.2 : Launch Optional Streamlit Dashboard
  - Activity 6.3 : Validate Outputs and Edge Cases
- Conclusion: 50%

1 2 3 4 5 6

Multimodal Attendance & Engagement Agent

AR Assigned to **Arpan Mishra**

Description

The **Multimodal Attendance & Engagement Agent** successfully demonstrates the practical application of multimodal artificial intelligence techniques in educational environments. By integrating computer vision, audio signal processing, and rule-based agent reasoning, the system moves beyond traditional attendance mechanisms to provide deeper insights into classroom participation and engagement dynamics.

Throughout the project, a milestone-driven development approach was followed to ensure systematic implementation and validation of each component. The Vision Module enabled reliable detection and tracking of participants, while presence-based logic ensured accurate attendance marking. The Audio Module added contextual awareness by analyzing noise levels and speech activity without compromising privacy. These multimodal signals were effectively fused by the Multimodal Agent, enabling interpretable engagement scoring and trend analysis.

The Analytics and Recommendation Engine transformed raw engagement data into meaningful insights, offering context-aware teaching suggestions for instructors and reflective session analytics for students. These outputs were designed as decision-support tools, maintaining ethical boundaries by avoiding automated judgments or evaluations. Comprehensive testing and validation confirmed the system's robustness under real-world conditions, including variations in lighting, noise, and participant movement.

Overall, this project highlights how multimodal AI systems can enhance classroom awareness while preserving transparency, privacy, and human oversight. The modular architecture ensures scalability and adaptability, making the system suitable for deployment in classrooms, training sessions, and workshops. The Multimodal Attendance & Engagement Agent serves as a strong foundation for future enhancements and demonstrates the potential of intelligent, ethical AI solutions in modern education.

Resources

No resources available

Project Progress

50%

Add Demo Link | Add Github Link

- Pre-requisites 50%
- Project Workflow 50%
- EPIC 1: Requirements Specification and Environment Setup 50%
- EPIC 2: Vision Module Development 50%
- EPIC 3: Audio Module Development 50%
- EPIC 4: AI IMPLEMENTATION 50%
- EPIC 5 : Analytics and Recommendation Engine 50%
- EPIC 6 : Interface, Testing, and Deployment 50%
- Conclusion: 50%

Conclusion: